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SUPERSEDING
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MILITARY SPECIFICATION

KIT, MAINTENANCE, ELECTRICAL-ELECTRONIC; CABLE AND CABLE HARNESS

This specification has been approved by the Naval Air Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers an electrical-electronic maintenance kit containing consumable components, tools and equipment designed for use by shore based and shipboard personnel, to maintain, modify and repair electronic cables and cable harnesses.

1.2 Classification.

1.2.1 Components case (see 3.5.2). Components case shall contain the items listed in Table I.

1.2.2 Tool case (see 3.5.4). The tool case shall contain the items listed in Table II.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

Beneficial comments, (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to the Naval Air Engineering Center, Systems Engineering and Standardization Department (Code 93), Lakehurst, NJ 08733-5100, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4940

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SPECIFICATIONS

Federal

FF-P-101	Padlocks.
QQ-A-250/8	Aluminum Alloy, 5052, Plate and Sheet.
QQ-N-290	Nickel Plating (Electrodeposited).
QQ-S-698	Steel, Sheet and Strip, Low Carbon.
QQ-S-763	Steel bars, Wire, Shapes, and Forgings, Corrosion-Resistant.

Military

MIL-P-116	Preservation, Methods of.
MIL-C-4150	Case, Carrying, Water - Vapor Proof.
MIL-S-5059	Steel, Corrosion-Resistant (18-8), Plate, Sheet and Strip (Asg.).
MIL-W-5846	Wires, Electrical, Chromel and Alumel, Thermocouple
MIL-P-15024	Plates, Tags and Bands for Identification of Equipment.
MIL-C-28777/5	Cable Assembly, Electrical Test Equipment, (3 wire, 3 prong, 125 volt, Molded) Grounding Plug Connector, 60 Hertz.
MIL-M-38784	Manuals, Technical, General Requirements for Preparation of.
MIL-S-81824	Splice, Electric, Permanent, Crimp Style, Copper, Insulated, Environment Resistant
MIL-T-83507/14	Tool Kits, Eletrical Connectors, Contacts, Connector and Cable Accessories, Tool Kit, General Purpose (Air/Nitrogen Heat Gun).

STANDARDS

Federal

FED-STD-595	Colors.

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STANDARDS (cont'd)

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109	Quality Assurance Terms and Definitions.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-130	Identification Marking of U.S. Military Property.
MIL-STD-454	Standard General Requirements for Electronic Equipment.
MIL-STD-783	Legends for Use in Aircrew Stations and on Airborne Equipment.
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of.
MIL-STD-810	Environmental Test Methods.
MIL-STD-889	Dissimilar Metals.
MS3106	Connector, Plug, Electric, Straight.

Supplement

See Supplement 1 for list of associated specification sheets.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herien. Unless otherwise specified, the issue of the documents which are DOD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

ASTM A269	Specification for Seamless and Welded Austenitic Stainless Steel Tubing for General Service.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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2.3 Order of precedence. In the event of a conflict between the text of this specification and the reference cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with applicable specification sheets. In the event of conflict between the requirements of the basic specification and the specification sheet, the requirements of the specification sheet shall prevail.

3.2 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.2.1).

3.3 Materials and components. Materials and components shall conform to the applicable specifications, and be as specified herein. Materials which are not covered by specifications or standards shall be of the best commercial quality, the lightest practical weight, and entirely suitable for the purpose intended.

3.3.1 Corrosion resistance. Materials shall be corrosion-resistant or treated to resist corrosion by fuel, oil or salt spray.

3.3.2 Fungus-inert materials. Materials shall conform to the fungus-inert requirements specified in MIL-STD-454, Requirement 4.

3.3.3 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.3.4 Protective treatment. Materials that are subject to deterioration, when exposed to environmental conditions likely to occur during service usage, shall be protected against such deterioration in a manner that will not prevent compliance with the performance requirements of this specification.

3.4 Figures. Except for the dimensions shown, Figures 1 through 12 are descriptive and for convenience of identification. They are not intended to exclude items which otherwise conform to this specification.

3.5 Construction.

3.5.1 Cases. With the exception of the stacking provisions, handle requirements, tamper-proof seal, and the lifting ring and wire, the cases shall conform to the requirements of MIL-C-4150 (fiberglass construction). Color shall be in accordance with FED-STD-595; Color No. 13538, 23538 or 33538.

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3.5.1.1 Handle. A single handle shall be attached to each case. It shall be symmetrically located with respect to weight distribution of the loaded case. Each case shall be subjected to the handle test specified in 4.6.8.

3.5.1.2 Manual pressure relief valve. The components case shall be equipped with a manual pressure relief valve capable of being actuated without the use of tools.

3.5.1.3 Hasp. Each case shall be equipped with a hasp that is compatible with a padlock conforming to the requirements of FF-P-101, Type ED, 2 Inch Size.

3.5.2 Components case. The dimensions and configuration of the components case shall be as indicated in Figure 1. The components listed in Table 1 shall be stowed within the case. The hinges for the case shall be such that the top half of the case will become detached from the bottom half when the case is opened to a 150° angle ±10°.

3.5.2.1 Identification plates. The components case shall be equipped with a hinged aluminum alloy identification plate (see Figure 3). The plate shall be 0.090 ±.009 inch thick. It shall delineate the components and their locations. The marking shall be in accordance with MIL-P-15024, Type H.

3.5.2.2 Weight. The weight of components case, containing the components listed in Table 1, shall not exceed 50 pounds.

3.5.2.3 Tubing rack. Items 12 through 28 and Item 32 shall be stowed on individual plastic spools in a rack that can be removed from its recess in the components case.

3.5.3 Covers - individual compartments. Each compartment shown in Figure 1 (tray insert, top) shall be equipped with a hinged, springloaded, transparent, plastic cover. The plastic covers shall be 0.125 inch thick (minimum).

3.5.4 Tool case. The dimensions of the tool case shall be as indicated in Figure 2. Details of the tool case shall be as indicated in Figure 4.

3.5.4.1 Weight. The weight of the tool case, containing the tools listed in Table II, shall not exceed 40 pounds.

3.5.4.2 Instruction plate. The tool case shall be equipped with a hinged aluminum alloy plate (see Figure 4). The plate shall be 0.090 ±.009 inch thick. It shall contain operating instructions for the tools and accessories contained in the tool case. The marking shall be in accordance with MIL-P-15024, type H.

3.5.4.3 Filter drain. The tool case shall be equipped with a drain opening which directs moisture purged from the compressed air heating unit filter to the outside of the tool case. The leakage requirement of MIL-C-4150 shall not be applicable to the tool case, and the pressure relief valve shall not be required.

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3.5.5 Tools. The tools listed in Table II shall be stowed within the tool case as illustrated in Figure 4.

3.5.5.1 Maintainability. The tools shall be so constructed that all parts (specifically the heating elements) subject to replacement during the normal life of the assembly, are easily removable without the necessity of special tools or of returning to the manufacturer.

3.5.6 Heat gun with motorized impeller. The heat gun, with motorized impeller, shall be equipped with a three position switch (hot-cold-off). It shall be capable of operating at 115 VAC ±5%, 60 Hz ±5%, 15 amp (max.) After a warm up time of one minute the temperature at the gun nozzle shall be 608°F ±20%. It shall have a nominal air flow of 10 standard cubic feet per minute (SCFM). The nozzle shield shall be equipped with three detents (120° apart) to position and retain the reflectors (see 3.5.6.3) during normal usage. A decal shall be affixed to the gun with the following words inscribed thereon: "DO NOT USE ON AIRCRAFT". The preferred configuration of the gun is shown in Figure 5 (see 4.6.6 and 4.6.9.1).

3.5.6.1 Electrical cord. The electrical cord and plug shall be a 3-wire, 3-prong molded plug and wire assembly in accordance with MIL-C-28777/5-1 or commercial equivalent with UL type SJO (3/16) wire. The length shall be 6 ±0.5 feet.

3.5.6.2 Heat gun base. The kit shall be equipped with a heat gun base. The preferred configuration is shown in Figures 5. The base shall insure that the gun remains in a stationary position during use.

3.5.6.3 Heat gun reflectors. The kit shall be equipped with four heat gun reflectors, fabricated from cold rolled steel conforming to the requirements of QQ-S-698. The reflectors shall be 0.020 ±0.005 thick and nickel plated in accordance with QQ-N-290, Class 1, Grade B. The preferred configurations of the reflectors are shown in Figures 6, 7, 8, and 9. The design shall insure that the air is directed uniformly around the component to be heated.

3.5.7 Compressed air heating unit. The kit shall be equipped with a compressed air heating unit. The unit shall operate at 115 VAC ±5%, 50-400 Hz ±5%, 1 phase, 5 amp (max). Within a period of 3 minutes the unit shall be capable of delivering air at 920±20°F with the temperature control set at maximum. It shall operate at 50 psig with a flow of approximately 2.5 SCFM. The preferred configuration is shown in Figure 10. The unit shall not leak or malfunction when operated from a 200 psig pressure source. It shall be subjected to the test specified in 4.6.4 and in 4.6.9.3.

3.5.7.1 Components. The unit shall include, but not be limited to the following components:

Heat gun - non-sparking (3.5.7.2)
Reflectors (3.5.7.3)
Air hose adapter (3.5.7.4)
Air hose assembly (3.5.7.5)
Module (3.5.7.6)

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3.5.7.2 Heat gun - non-sparking. The non sparking heat gun shall accept air from a compressed air system and channel it over an electrical heating element. With the exception of the area forward of the heat shield, the gun shall be capable of being hand held without the use of gloves or other heat protection devices. The heating element shall be field replaceable without the use of special tools. The weight of the gun shall not exceed 16 ounces (see 4.6.2.1).

3.5.7.3 Reflectors. The compressed air heating unit shall be equipped with 5 heat reflectors. The reflectors shall be fabricated from corrosion resistant steel conforming to the requirement of MIL-S-5059, Type 302 or 304 CRES per ASTM A269 or 304 CRES per QQ-S-763. The preferred configurations of the reflectors are shown in Figures 11.

3.5.7.4 Air hose adapter. An air hose adapter shall be provided which has a quick-disconnect female coupler capable of connecting to the module air fitting and which also has a quick-disconnect male fitting conforming to Parker-Hannifon Tru-Flate model 2C-1/4 NPT, or equivalent.

3.5.7.5 Air hose assembly. The air hose assembly shall be 20± 0.5 feet long and consist of an air/nitrogen supply hose, electrical wires, and a flexible, protective sheeth, power wire, 18 gauge. The cord shall be of Type SPT-2, Thermocouple wires shall be in accordance with MIL-W-5846.

3.5.7.6 Module. The module shall be secured in the tool case and be capable of being removed from the case within 30 seconds (See figure 4).

3.5.7.6.1 Fabrication. The front and top of the module shall be fabricated from 0.125 (nominal) inch thick aluminum alloy conforming to the requirement of QQ-A-250/8. Only the controls and indicators required for normal use shall be exposed. The rear cover shall be fabricated from molded acrylic - PVC thermoplastic.

3.5.7.6.2 Panel protection. The panel, indicators and controls shall be protected by recessing the panel within the case

3.5.7.6.3 Components The following components shall be permanently installed in the module.

Pressure regulators (3.5.7.6.4)
Pressure gauge (3.5.7.6.5)
Pressure switch (3.5.7.6.6)
Air filter (3.5.7.6.7)
Air inlet fitting (3.5.7.6.8)
Electrical switch (3.5.7.6.9)
Electrical cord (3.5.7.6.10)
Indicator lights (3.5.7.6.11)
Temperature controller (3.5.7.6.12)

3.5.7.6.4 Pressure regulator. The pressure regulator shall be capable of regulating the air pressure within the range 0-100 psig.

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3.5.7.6.5 Pressure gauge. The pressure gauge shall have a range from 0 to 30 psig with an accuracy of ±10% over the entire range.

3.5.7.6.6 Pressure switch. The pressure switch shall be preset to open the electrical circuit when the air pressure in the unit is below 4.75 ±0.5 psig (see 4.6.5 and 4.6.9.4).

3.5.7.6.7 Air filter. The air filter shall be a five micron absolute filter.

3.5.7.6.8 Air inlet fitting. The module shall be equipped with a high pressure, male, quick-disconnet fitting that is compatible with the air hose assembly. The fitting shall be equipped with a dust cover.

3.5.7.6.9 Electrical switch. The module shall include an on-off switch with a built-in circuit breaker that is compatible with 3.5.7.

3.5.7.6.10 Electrical cord. The electrical cord and plug shall be a 3-wire 3 prong, molded plug and wire assembly UL type SJO 3-18. The length shall be 6.0 ±0.5 feet. An adhesive back warning label (international orange background with black lettering) shall be attached directly behind the electrical three prong plug and shall read on both sides of the label as follows:

"HT900B (115 VAC, 50-400Hz, 1 phase, 7A). The attached standard universal power plug is not approved for use in hazardous locations as defined in NFPA 70-1984, article 500. If this heating device must be used in a hazardous location, as defined in NFPA 70-1984, article 500, the universal power plug must be replaced with a plug designed for use in a hazardous location. It is the responsibility of the user to provide appropriate connection to the power source. Failure to comply with these instructions may result in fire or explosion."

3.5.7.6.11 Indicator lights. The module shall include two press-to-test indicator lights conforming to the requirements of MIL-STD-454, requirement 50. One will illuminate when electrical power is applied to the module and one will illuminate when electrical power is applied to the heat gun.

3.5.7.6.12 Temperature controller. The temperature controller shall be capable of controlling the output air temperature within the range of approximately 700°F to 900°F.

3.5.7.6.13 Panel markings. The visible surface adjacent to panel facilities, such as controls, indicator lights, jacks, sockets, and fuse holders shall be marked with a suitable word, phrase, or abbreviation, indicating the use or purpose of the part. The markings shall be permanent, such as produced by engraving, etching, silk screening, or metal stamping, either directly on the panel or on attached metal plates, shall be legible and shall be of contrasting colors. When metal plates are used, they shall be attached in such manner that they will remain fixed in their original position. Abbreviation shall be in accordance with MIL-STD-783. Operating controls shall be provided with instructions which will permit the operators to set the control easily and correctly.

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3.6 Crimp tool. The crimp tool shall be as specified on the manufacturer's control drawing listed in QPL-81824 and for splices conforming to MIL-S-81824 as shown in Figure 12.

3.7 Interchangeability. Interchangeability shall be in accordance with MIL-STD-454, Requirement 7.

3.8 Identification of product. The kit shall be marked for identification in accordance with MIL-STD-454, Requirement 67.

3.8.1 Nameplate. A metal nameplate shall be permanently secured to the top surface of each case. Identification marking shall be in accordance with MIL-STD-130.

3.8.2 Size and weight. The weight of the loaded case and cubic dimensions shall be marked on three external surfaces of each case.

3.9 Safety (personnel hazard). The kit, including the tools, shall conform to the safety (personnel hazard) requirements specified in MIL-STD-454, Requirement 1.

3.10 Workmanship. The general appearance of the kit shall reflect the highest standard of workmanship and uniformity of quality. In addition, the kit shall be free from contamination, foreign matter and damage such as dents, cracks, or deformation which could adversely affect the safety, performance, reliabilty, or durability and use of the kit. The kit shall conform to MIL-STD-454, Requirement 9.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet the requirements of section 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractors of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Inspection terms. MIL-STD-109 shall govern the definition of terms used herein.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

First article inspection (see 4.3).

Quality conformance inspection (see 4.4).

(a) First article inspecton. First article inspection consists of examinations and tests performed on samples (which are representative of the production item) after the award of a contract. The examinations and tests are conducted to determine that the kits meet the requirements of this specification (See 4.3).

(b) Quality conformance inspection. Quality conformance inspection consists of examinations and tests performed on the production items to determine that the kits meet the requirements of this specification (See 4.4).

4.3 First article inspection. First article inspection shall consist of all the inspections of this specification.

4.3.1 First article samples. Unless otherwise specified, as soon as practicable after award of a contract or order, the manufacturer shall submit samples as specified in the contract or order. The samples shall be representative of the construction, workmanship, components, and materials to be used during production.

4.4 Quality conformance inspection. The quality conformance inspection shall consist of the following examinations and tests.

Visual examination (4.6.1)

Functional tests (4.6.9)

4.4.1 Sampling for tests and examinations of kits. Sampling shall be in accordance with MIL-STD-105, inspection level III, acceptable quality level 0.40.

4.5 Test conditions.

4.5.1 Temperature and pressure. Unless otherwise specified, tests shall be conducted at an ambient temperature of approximately 77°F ±5° and a barometric pressure of approximately 29.92 in. Hg. Components (and tools) shall be housed at the ambient temperature for 1 hour prior to test.

4.6 Inspection methods.

4.6.1 Visual examination.

4.6.1.1 Examination of samples. The sample kits shall be examined to determine conformance with the requirements of this specification, including those requirements not covered by specific tests.

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4.6.1.2 Examination of preparation for delivery. The packaging and packing of the kits shall be examined to ascertain that the preparation for delivery conforms to Section 5 of this specification.

4.6.1.3 Examination of contents of kits. The contents of the components case and tool case shall be examined for conformance to Table I and Table II.

4.6.2 Explosive atmosphere test (compressed air heating unit). The explosive atmosphere test shall consist of two separate tests; a heat gun test and a module test.

4.6.2.1 Heat gun (non-sparking) test. The non sparking heat gun test shall be conducted in accordance with MIL-STD-810, Method 511.1, Procedure 1. The following details shall apply:

(a) Connect the compressed air heating unit to an electric power source of 115 VAC, 60 or 400 Hz and to a compressed air or nitrogen source of 80 to 200 psig.

(b) Attach a termination sleeve reflector to the heat gun nozzle and position the heat gun inside the explosive chamber.

(c) The chamber must be vented and an explosive atmosphere continously supplied to the chamber during testing.

(d) The test shall be continued for 5 minutes minimum, after which the power shall be turned off, the air flow reduced to zero, and the explosive atmosphere in the chamber tested for correct fuel-to-air ratio. This step shall be repeated three times.

(e) Continue operation in the chamber for 3 minutes. Then turn the switch in the handle to the off position for 1 minute. Repeat three times.

(f) Continue operation in the chamber for 3 minutes. Then pinch the hose assembly outside the chamber to shut off the air for 1 minute. Repeat three times.

(g) Repeat (a) through (f) at 5000, 10,000 and 15,000 foot simulated attitudes.

4.6.2.2 Module test. The module test shall be conducted in accordance with MIL-STD-810, Method 511.1, Procedure 1. The following details shall apply:

(a) Connect the compressed air heating unit to an electric power source of 115 VAC, 60 or 400 Hz, and to a compressed air or nitrogen source of 80 to 200 psig.

(b) Position the control module inside the explosion chamber so that the on-off power switch can be actuated from outside the chamber.

(c) Set the pressure regulator for 5-7 psig, and do not readjust.

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(d) The on-off switch shall be actuated three times at each simulated altitude (sea level, 5000, 10,000 and 15,000 feet), while the explosive atmosphere is maintained at 160 ±5°F.

(e) Disconnect the air or nitrogen supply and repeat step (d).

4.6.3 Compressed air heating unit air pressure test. The compressed air heating unit air pressure test shall be conducted as follows:

(a) Insert the reflector shown in Figure 11A into the heating unit.

(b) Connect the unit to an air source.

(c) Increase the air source pressure to 200 psig.

(d) Turn the pressure regulator knob (clockwise direction) until the regulator is in the full open position.

(e) The unit shall be subjected to the above pressure for a period of three minutes. Upon completion of the test there shall be no evidence of rupture, distortion or damage to the unit.

4.6.4 Compressed air heating unit performance test. The performance test shall consist of 3 separate tests; each with a different parameter.

4.6.4.1 Test 1.

(a) Connect the heating unit to an electrical power source having 109 VAC, 60 or 400 Hz ±5% and an air pressure of 90 to 200 psig.

(b) Insert reflector shown in Figure 11D.

(c) Splice two lengths of number 20 stranded, tinned copper electrical wire.

(d) Position solder sleeve terminator (Part No. M81786/8-1) over the wires so that the solder insert is positioned around the splice.

(e) Adjust the air pressure regulator to 5-7 psig.

(f) Move the power switch to the on position and set the temperature control at the maximum setting. Do not readjust the pressure regulators.

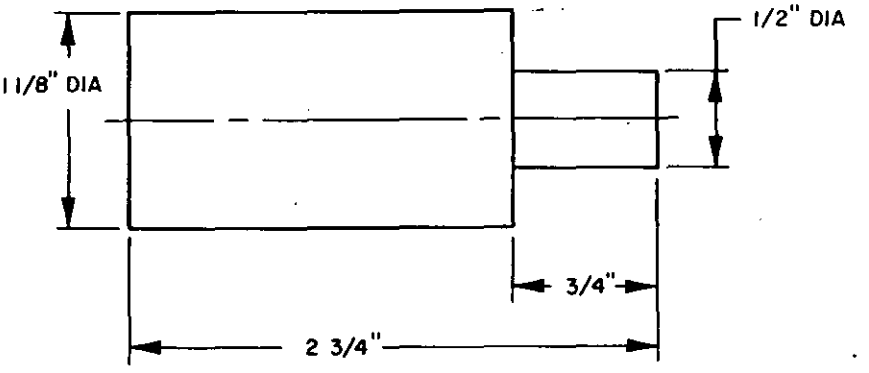
(g) Apply heat to the solder sleeve terminator.

(h) The solder shall melt and the tubing shall shrink to the size of the insulation within 40 seconds. With the exception of the flux residue removal procedure, the solder joint shall meet requirement 5 of MIL-STD-454.

(i) Replace reflector shown in Figure 11D with reflector shown in Figure 11B and continue as follows.

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(j) Position molded component (Part No. M81786/6-5) over a wood dowel of the dimensions specified below.



(k) Apply heat to the molded component.

(l) The component shall shrink to the size of the wood dowel within 3 minutes.

4.6.4.2 Test 2. This test shall be conducted using connectors MS3106E22-14P with 5 lengths of number 18 stranded wire soldered to pins J-T-V-P-C. A half inch length of tubing Part No. M81786/2-4 shall be placed over each soldered connection.

(a) Insert reflector shown in Figure 11C.

(b) Connect the heating unit to an electrical power source having 115 VAC, 60 or 400 Hz ±5%.

(c) Set the air pressure to 5-7 psig.

(d) Turn on the electrical power and set the temperature control at the minimum setting. Do not readjust the pressure regulator.

(e) Apply heat to the tubing on solder cup "T".

(f) The tubing shall shrink to the solder cup and wire insulation within 10 seconds. The adjacent tubing shall show no signs of shrinking and none of the soldered terminations shall show signs of melting during the test.

(g) Repeat steps (e) and (f) for solder cups J, V, P, and C.

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4.6.4.3 Test 3.

(a) Connect the heating unit to an electrical power source having 109 VAC, 60 or 400 Hz ±5% and an air source pressure of 90 to 200 psig.

(b) Insert reflectors shown in Figure 11E.

(c) Place a 4-inch length of tubing (Part No. M81786/3-9) on a length of 9/16 inch diameter wood dowel.

(d) Adjust the air pressure regulator to 5-7 psig.

(e) Move the power switch to the on position and set the temperature control at the maximum setting. Do not readjust the pressure regulator.

(f) Apply heat to the tubing.

(g) The tubing shall shrink to the size of the wood dowel within 60 seconds.

4.6.5 Pressure switch test. Upon completion of the previous test (4.6.4.3), reduce the air pressure until the power indicator light indicates electrical power has been removed from the heating unit. The pressure gauge shall indicate 4.0 max psig within 3 minutes.

4.6.6 Heat gun (with motorized impeller) performance test.

4.6.6.1 Test conditions. The heat gun shall be connected to an electrical power source of 109 VAC, 60 Hz.

4.6.6.2 The test shall be conducted as follows:

(a) Insert reflector shown in Figure 6 into the heat gun.

(b) Place a 4 inch length of tubing (Part No. M81786/3-6) on a length of 3/16 inch wood dowel.

(c) Place switch in the "hot" position and apply heat to the tubing. The tubing shall shrink to the size of the wood dowel within 40 seconds.

4.6.6.3 Repeat 4.6.2.2 substituing the reflector shown in Figure 7, a 9/16 inch diameter wood dowel, and a 4 inch length of tubing (Part No. M81786/3-9). The tubing shall shrink to the size of the wood dowel within 65 seconds.

4.6.6.4 Continue the test as follows:

(a) Insert reflector as shown in Figure 8 into the heat gun.

(b) Splice two lengths of number 20 stranded, tinned, copper electrical wire.

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(c) Position solder sleeve (Part No. M81786/8-1) around the wires so that the solder insert is positioned around the splice.

(d) Apply heat to the solder sleeve.

The solder shall melt and the tubing shall shrink to the size of the insulation within 40 seconds. With the exception of the flux residue removal provision, the solder joint shall meet Requirement 5 of MIL-STD-454.

4.6.6.5 Repeat 4.6.6.4 substituting the reflector shown in Figure 9, number 16 stranded tinned copper electrical wire, and Part No. M81786/8-2. The solder shall melt and the tubing shall shrink to the size of the insulation within 50 seconds. With the exception of the flux residue removal provision, the solder joint shall meet Requirement 5 of MIL-STD-454.

4.6.6.6 Connect the heat gun to the power source having 121 VAC, 60 Hz and repeat the test specified in 4.6.6.2 through 4.6.6.5.

4.6.6.7 Cold switch. Immediately upon completion of the test specified in 4.6.6.6 place the switch in the "cold" position. After a period of 3 minutes it shall be established that unheated air is being emitted from the heat gun.

4.6.7 Drop test. The components case and the tool case shall be subjected to and pass the drop test specified in MIL-C-4150.

4.6.8 Handle test.

4.6.8.1 Components case. With the components case containing all the items listed in Table I the handle test shall be conducted as follows:

(a) Secure one end of a length of line, of sufficient strength, to the handle of the case.

(b) Secure the other end of the line to an overhead support so that the case is suspended 12 inches from the floor.

(c) Lift the case to a height of 4 feet above the floor and permit the case to drop free.

(d) Repeat step (c) 5 times.

The handle shall not separate or become loose from its mounting and shall show no evidence of distortion. In addition, there shall be no evidence of distortion or damage to the case.

4.6.8.2 Tool case. With the tool case containing all the items listed in Table II, repeat the test procedure specified in 4.6.8.1.

4.6.9 Functional tests.

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4.6.9.1 Heat gun with motorized impeller. The heat gun functional test shall be conducted as follows:

(a) Connect the heat gun to a 115 VAC, 60 Hz power source.

(b) Place switch to the "hot" position.

Within a period of one minute it shall be established that the temperature of the air being emitted from the nozzle of the heat gun is 320°C ±20%.

4.6.9.2 Place the switch in the "cold" position. After a period of 3 minutes it shall be established that air being emitted from the heat gun is at ambient temperature ±39°F.

4.6.9.3 Compressed air heating unit. The compressed air heating unit functional test shall be conducted as follows:

(a) Remove the reflector from the heating unit.

(b) Set the air pressure to 5-7 psig and connect the unit to a 115 VAC 400 Hz power source.

(c) Turn the power switch on and turn the temperture control to the maximum setting. Within a period of 3 minutes it shall be established that the temperature of the air at the nozzle tip is 900°F ±10%.

(d) Turn the temperature control to the minimum setting. Within a period of 3 minutes it shall be established that the temperature of the air at the nozzle tip is 550°F ±30°F.

4.6.9.4 Pressure switch test. Upon completion of the above test (4.6.9.3), reduce the air pressure until the power indicator light indicates electrical power has been removed from the heating unit. The pressure gauge shall indicate 4.75 ±.05 psig within 1 minute.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Procedures for packaging and packing shall be in accordance with MIL-STD-794, Level A or C as specified (see 6.2). Preservation shall be in accordance with MIL-P-116, Method 1C-1.

5.2 Marking. In addition to any special marking required in the contract or order (see 6.2), marking shall be in accordance with MIL-STD-129 and MIL-STD-454, Requirement 67.

6. NOTES

6.1 Intended use. The maintenance kit covered by this specification is intended for use at all maintenance activities on Naval ships and shore stations to maintain, modify and repair electrical and electronic cables and cable harnesses.

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6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Whether first article inspection is required and where the samples shall be delivered (see 4.3.1, 6.2.1 and 6.2.1.1).

(c) Selection of applicable level of preservation and packaging required (see 5.1).

(d) Special marking, if required (see 5.2).

6.2.1 First article. When a first article inspection is required, the items should be a first article sample. The first article should consist of one kit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.2.1.1 Whether first article inspection is required. When a contractor is in continuous production of the kits from contract to contract, consideration should be given to waive the first article inspection.

6.3 Data. For the information of Contractors and Contracting Officers, any applicable documents listed in Secton 2 of this specification, or referenced lower-tier documents need not be prepared for the Government unless specified in the contract or order. The data to be furnished shall be listed on DD Form 1423 (Contractor Data Requirements List), which shall be attached to and made a part of the contract or order. NAVAIR Form 4200/25 (Drawings, Lists, and Specifications Required) shall be attached where applicable.

6.3.1 Manuals. The instruction manual shall be written in accordance with MIL-M-38784, a modified by the contract.

6.4 Subject term (key word) listing.

Cable
Cable harness
Compressed air heating unit
Crimp tool
Electrical maintenance kit
Heat gun, non sparking
Heat gun reflector
Heat gun with motorized impeller

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6.5 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing Activity
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(Project No. 4940-N510)

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TABLE I

COMPONENTS

KIT ITEM NO.	QUANTITY	PART NO.
1	100	M81786/1-1
2	100	M81786/1-2
3	100	M81786/1-3
4	100	M81786/1-4
5	20	M81786/2-1
6	20	M81786/2-2
7	20	M81786/2-3
8	20	M81786/2-4
9	20	M81786/2-5
10	21	M81786/2-6
11	21	M81786/2-7
12	1	M81786/3-1
13	1	M81786/3-2
14	1	M81786/3-3
15	1	M81786/3-4
16	2	M81786/3-5
17	2	M81786/3-6
18	1	M81786/3-7
19	1	M81786/3-8
20	1	M81786/3-9
21	1	M81786/3-10
22	1	M81786/4-1
22	1	M81786/4-1
23	1	M81786/4-2
24	1	M81786/4-3
25	1	M81786/4-4
26	1	M81786/4-5
27	1	M81786/4-6
28	1	M81786/4-7
29	1	M81786/4-8
30	1	M81786/4-9
31	1	M81786/4-10
32	1	M81786/5-1
33	2	M81786/5-2
34	2	M81786/5-3
35	2	M81786/5-4
36	3	M81786/6-1
37	4	M81786/6-2

NOTE: Items 12 through 28 and item 32 shall be reeled on plastic spools which shall be stowed in a removable rack. This rack shall permit easy insertion and removal of the tubing spools.

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TABLE I (Continued)

COMPONENTS		
KIT ITEM NO.	QUANTITY	PART NO.
38	6	M81786/6-3
39	8	M81786/6-4
40	8	M81786/6-5
41	3	M81786/6-6
42	3	M81786/6-7
43	3	M81786/6-8
44	3	M81786/6-9
45	3	M81786/6-10
46	4	M81786/6-11
47	4	M81786/6-12
48	3	M81786/6-13
49	3	M81786/6-14
50	800	M81786/8-1
51	800	M81786/8-2
52	400	M81786/8-3
53	800	M81786/8-1
54	800	M81786/8-2
55	200	M81786/8-3
56	500	M81786/9-1
57	500	M81786/9-2
58	400	M81786/9-3
59	200	M81786/10-1
60	100	M81786/10-2
61	50	M81786/10-3
62	45	M81786/10-4
63	20	M81786/11-1

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TABLE II

TOOLS

KIT ITEM NO.	QUANTITY	DESCRIPTION
64	1	Heat Gun-with Motorized Impeller (See Figure 5)
65	1	Heat Gun Base (See Figure 5)
66	1	Heat Reflector (See Figure 6)
67	1	Heat Reflector (See Figure 7)
68	1	Heat Reflector (See Figure 8)
69	1	Heat Reflector (See Figure 9)
70	1	Heat Gun - Non sparking (see Figure 10
71	1	Heat Reflector (See Figure 11A)
72	1	Heat Reflector (See Figure 11B)
73	1	Heat Reflector (See Figure 11C)
74	1	Crimp Tool (See Figure 12B)
75	1	Heat Reflector (See Figure 11E)
76	1	Crimp Tool Gauge (See Figure 12A)
77	1	Heat Reflector (see Figure 11D)

NOTE:

1. The kit shall include two sets of motor brushes for heat gun with motorized impeller, as a standard item.

2. For arrangement of tools see figure 4.

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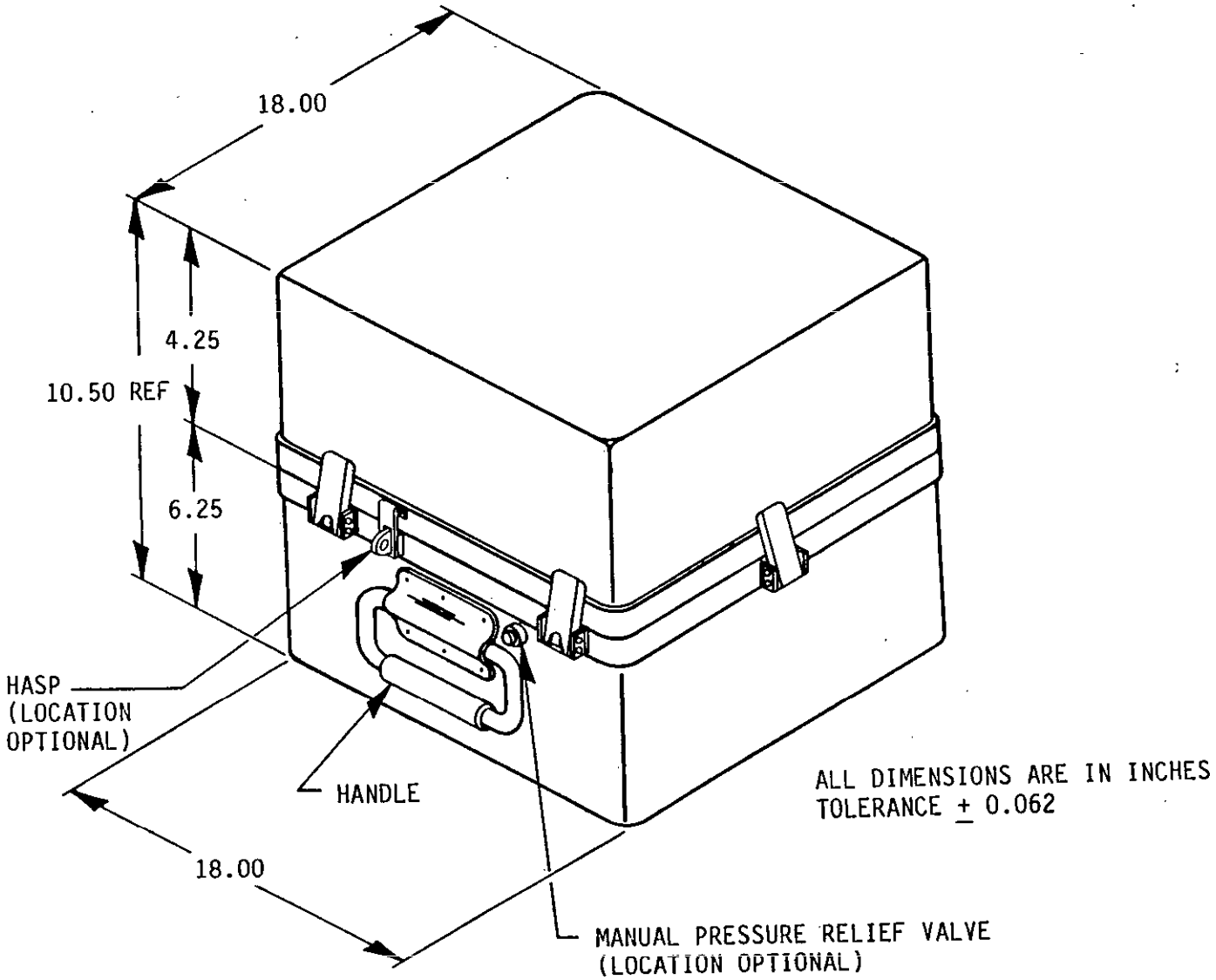
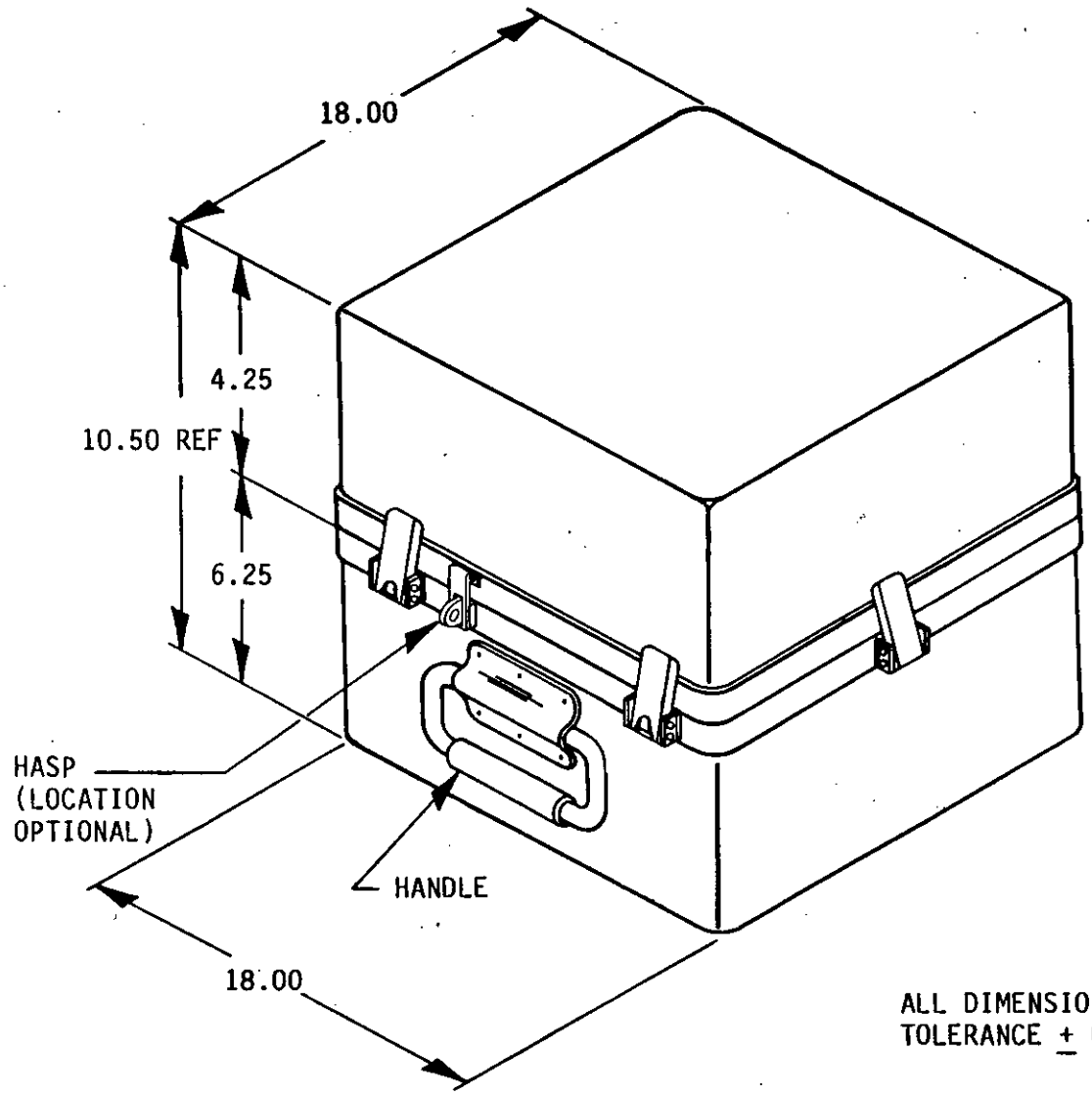


FIGURE 1. Components case.

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FIGURE 2. Tool case.

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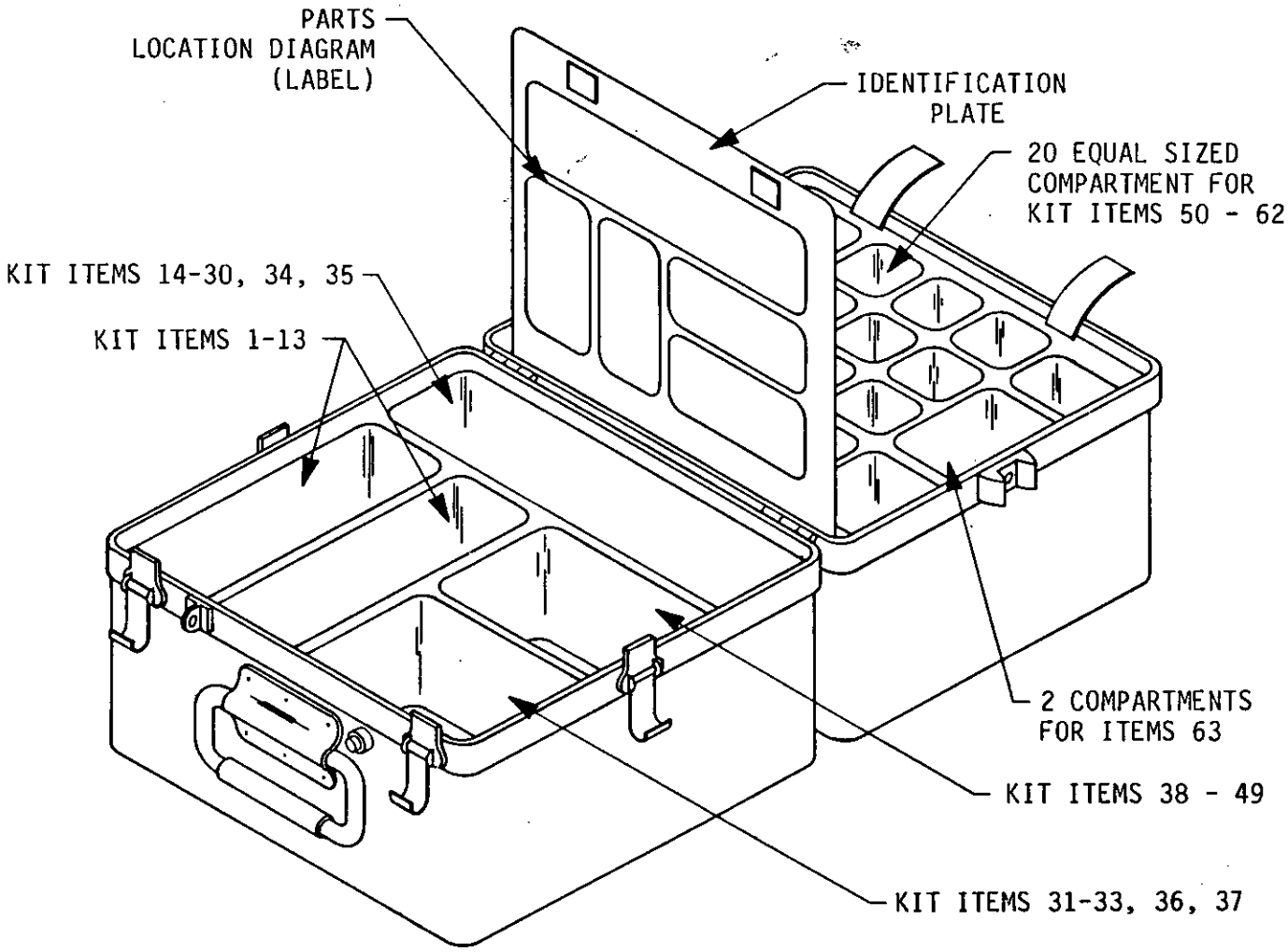


FIGURE 3. Arrangement of components.

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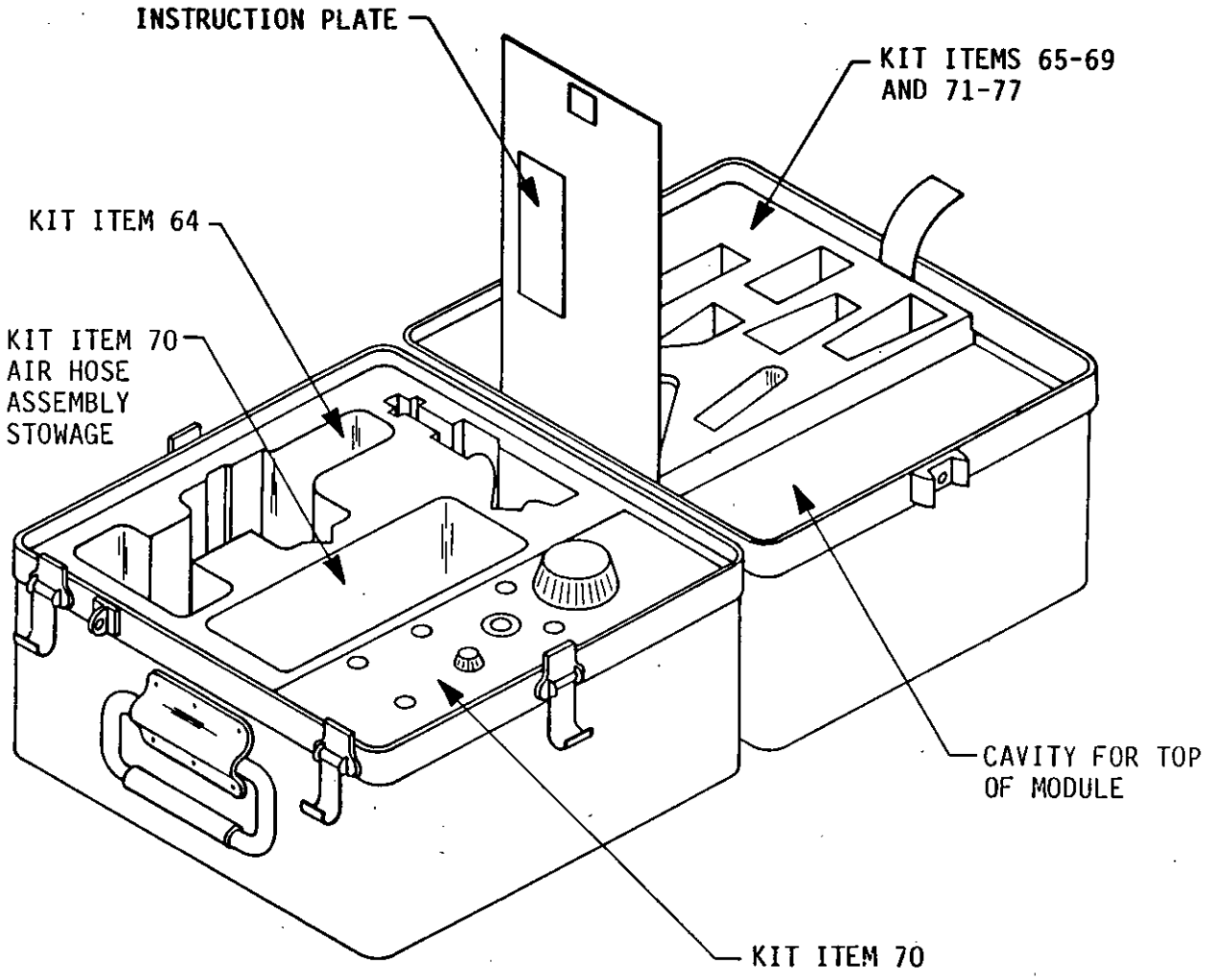


FIGURE 4. Arrangement of tools.

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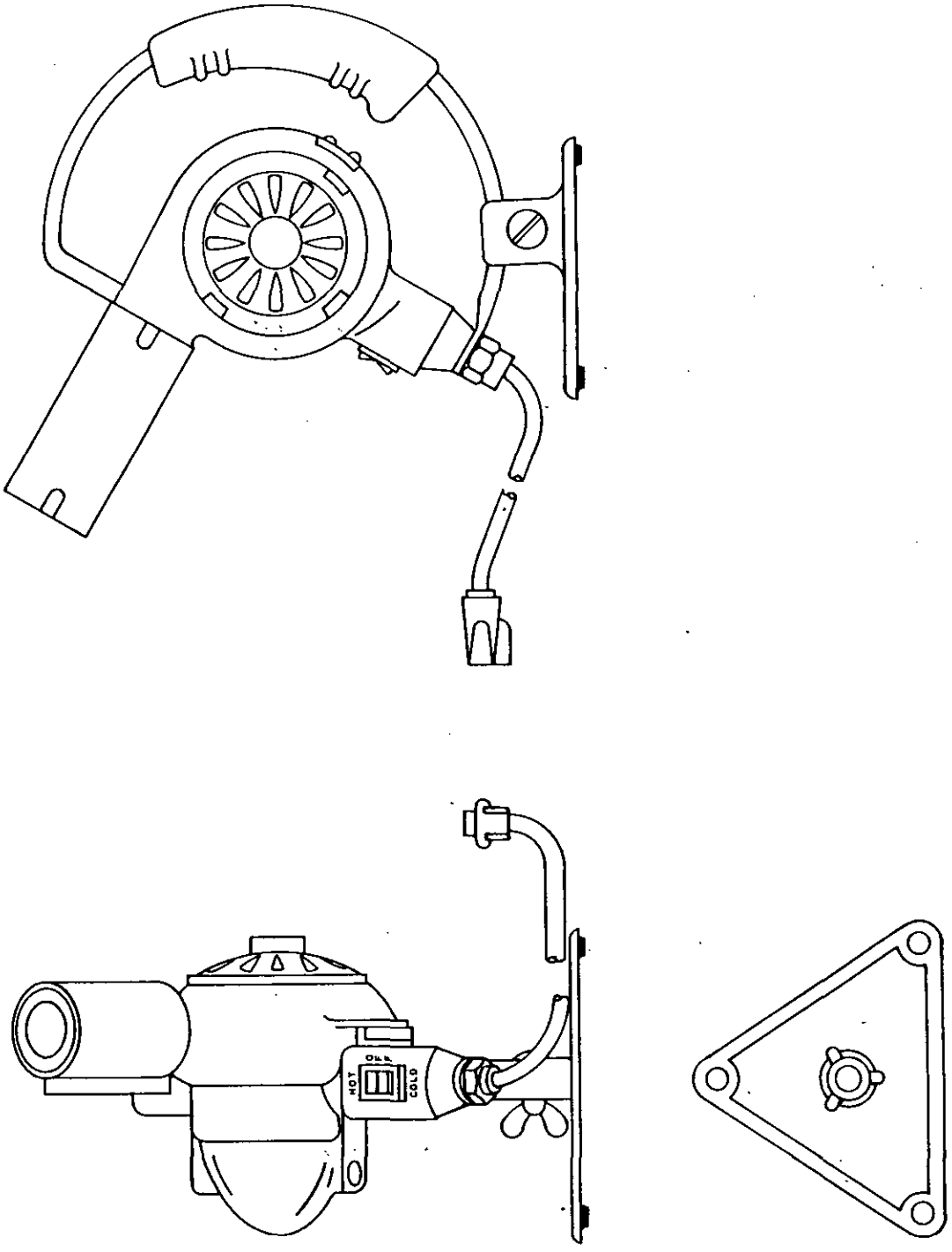
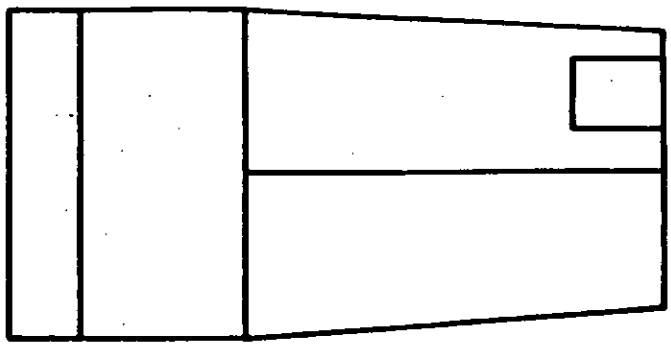
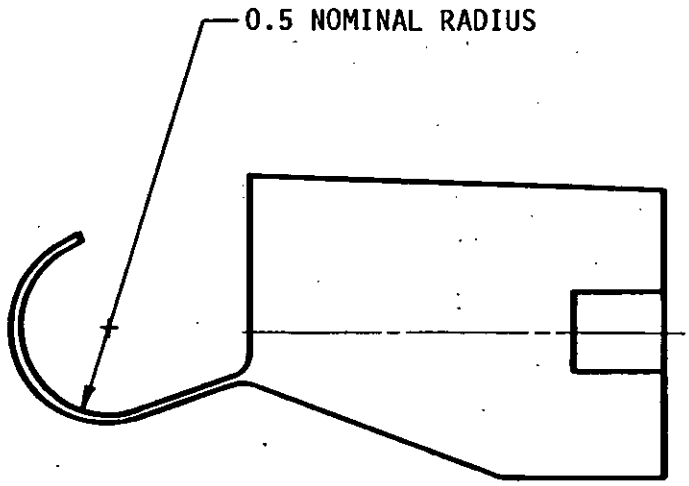


FIGURE 5. Heat gun (with motorized impeller) and base.

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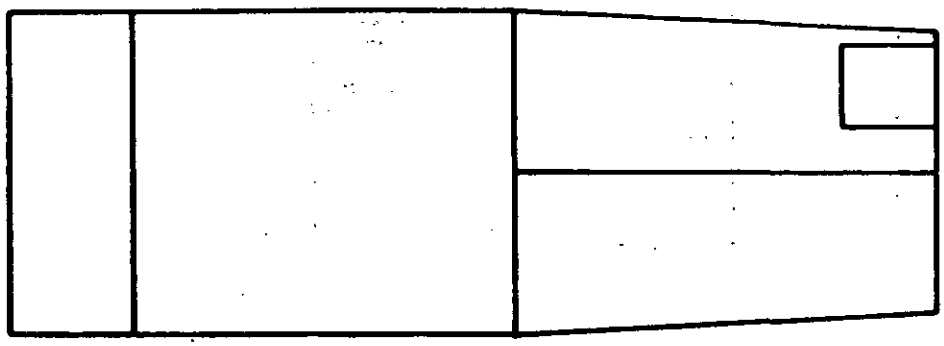


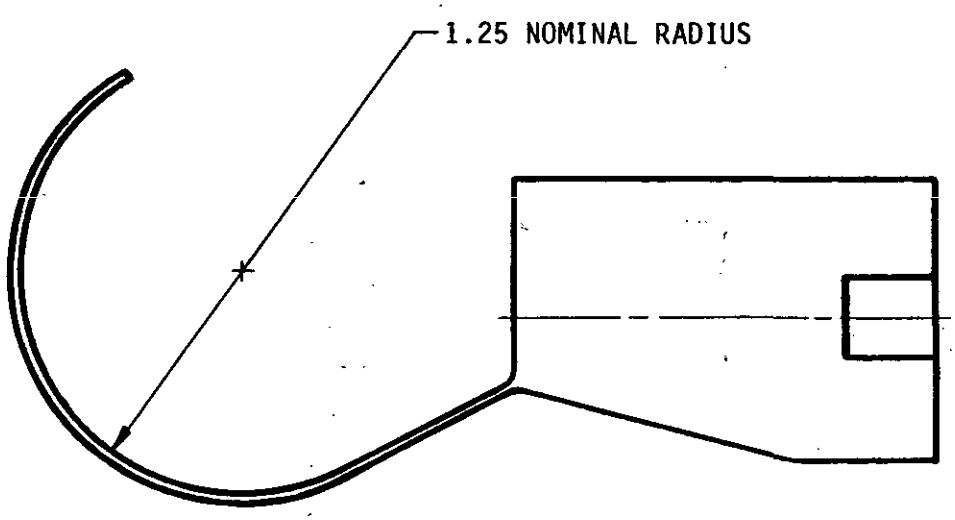


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FIGURE 6. Heat gun reflector.

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FIGURE 7. Heat gun reflector.

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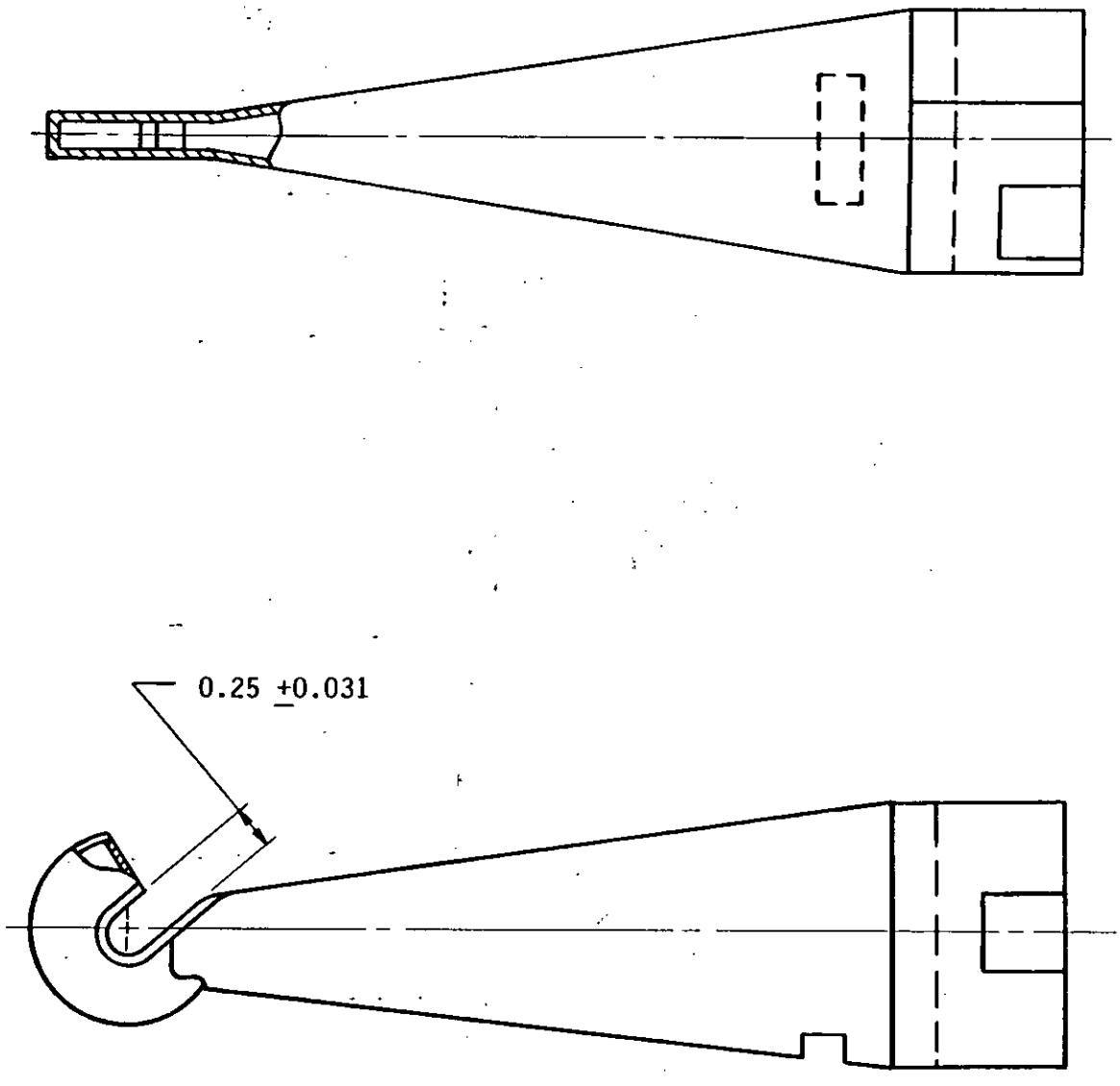
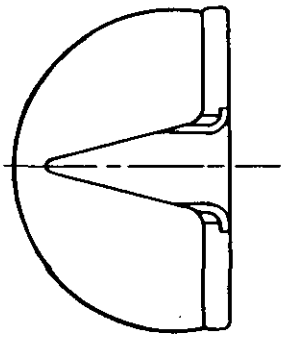
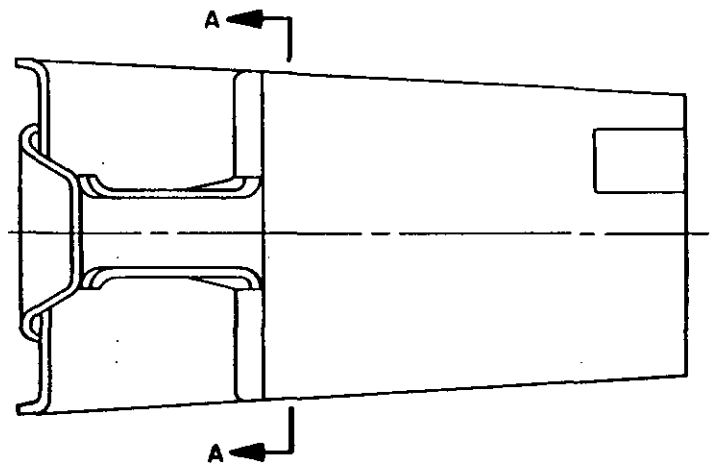


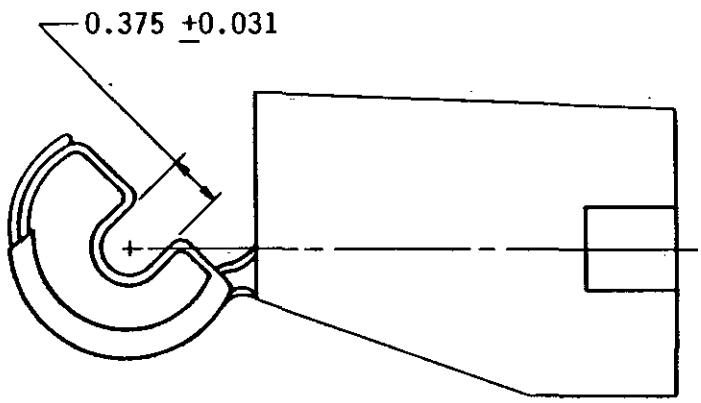
FIGURE 8. Heat gun reflector.

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SECTION A-A





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FIGURE 9. Heat gun reflector.

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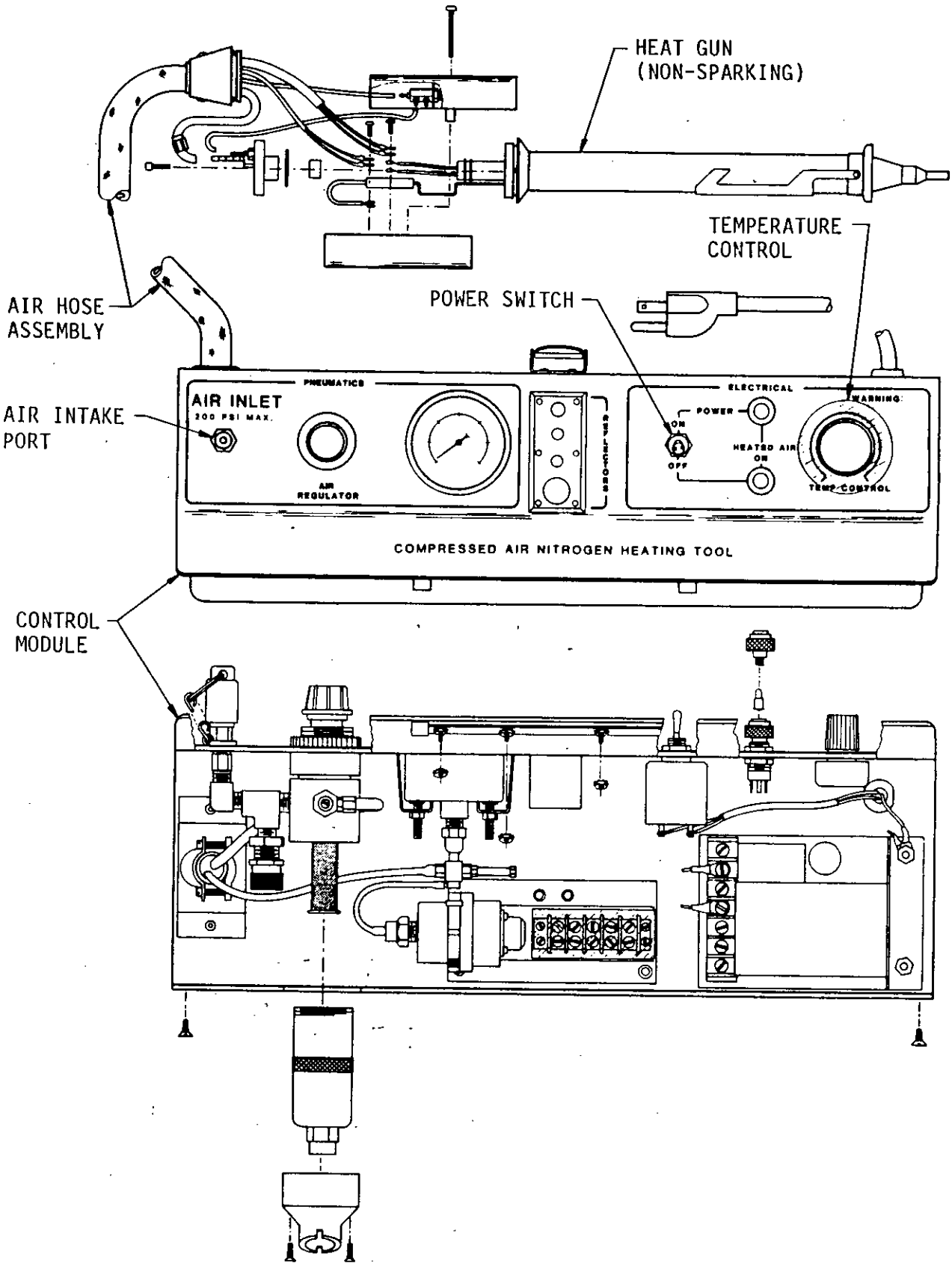


FIGURE 10. Compressed air heating unit.

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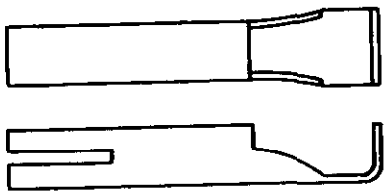


FIGURE 11A. For small termination sleeves and small diameter tubing.

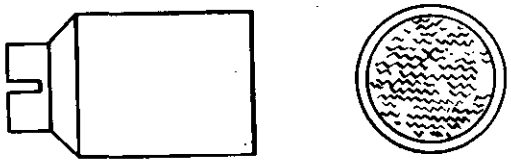


FIGURE 11B. For molded components and larger tubing.

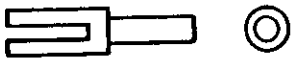


FIGURE 11C. For miniature tubing.

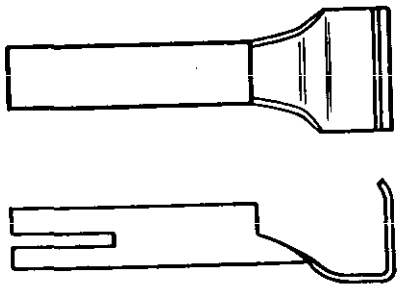


FIGURE 11D. For large termination sleeves and medium diameter tubing.

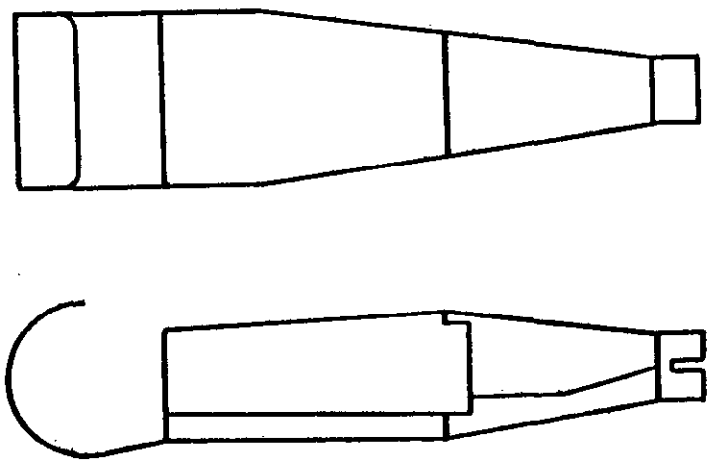


FIGURE 11E. For large molded components and tubing.

FIGURE 11. Compressed air heating unit reflector.

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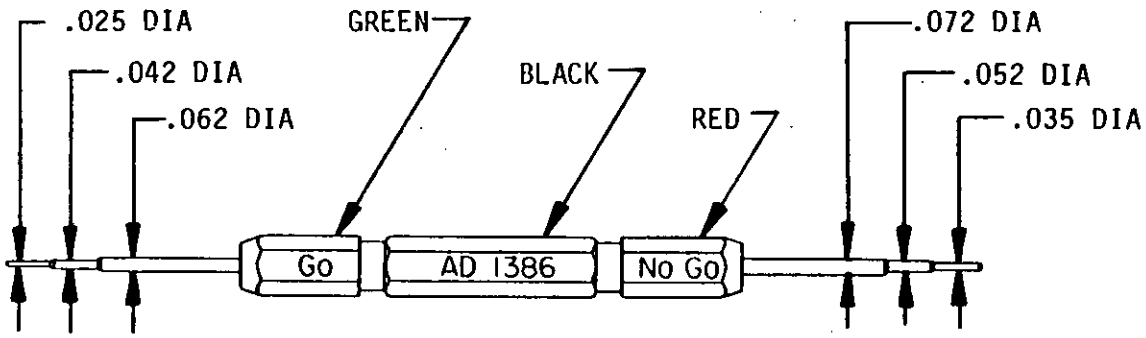


FIGURE 12A. Crimp tool gauge.

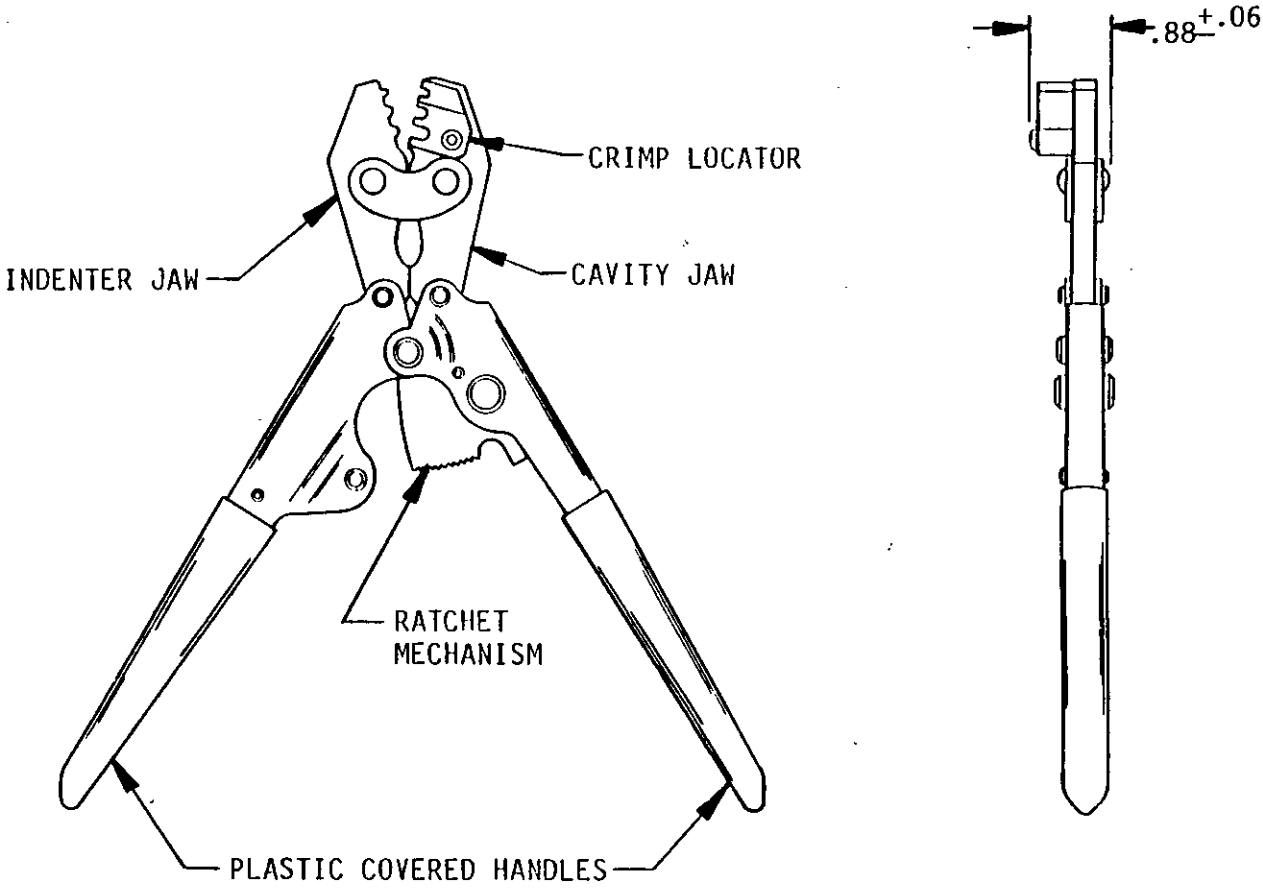


FIGURE 12B. Crimp tool.

FIGURE 12. Crimp tool and gauge.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-K-81786B	Kit, Maintenance, Electrical-Electronic; Cable and Cable Harnes

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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